8) Discuss the convergence or the Divergence of

$\{U_n\} = \frac{3n}{n+1}$

$\frac{\cancel{n}(3)}{\cancel{n}(1+\frac{1}{n})}$

$= \frac{3}{1}$

$\lim_{n\to\infty} = 3n \div n+1$

$= \frac{3\cancel{n}}{\cancel{n}} \div \frac{\cancel{n}}{\cancel{n}} + \frac{1}{n}$

$= 3 \div 1 + \frac{1}{n}$ $\therefore \frac{1}{n} = 0$

$= 3 \div 1 + 0$

$= \frac{3}{1}$

$= 3$ (Converges)

$\{U_n\} = \frac{n^2}{n^2+9}$

$\lim_{n\to\infty} \frac{n^2}{n^2+9}$

$= \frac{n^{\cancel{2}}}{\cancel{n}} \div \frac{n^{\cancel{2}}}{\cancel{n}} + \frac{9}{n}$

$= n \div n + \frac{9}{n}$ $\therefore \frac{1}{n} = 0$

$= \frac{\cancel{n}}{\cancel{n}}$

$= 1$ (converges)

$\{U_n\} = \frac{5n+3}{3n+2}$

$= \frac{5\cancel{n}}{\cancel{n}} + \frac{3}{n} \div \frac{3\cancel{n}}{\cancel{n}} + \frac{2}{n}$

$= 5 + 0 \div 3 + 0$

$= \frac{5}{3}$ (converges)

Q) $\{U_n\} = n^2$

$\lim_{n \to \infty} U_n$

$\lim_{n \to \infty} n^2 = \infty$

The given series goes to infinity, isn't bound

hence it diverges.

Q) $\{U_n\} = (-1)^n$

$\lim_{n \to \infty} U_n$

$\lim_{n \to \infty} (-1)^\infty = +\infty$ for even

$-\infty$ for odd

the series oscilats and goes to infinty,

hence it diverges

Q) $\{U_n\} = (-1)^n \cdot n$

$= +\infty$ for even

$-\infty$ for odd

given series is divergent.

Q) $\{U_n\} = n+1$

$\lim_{n \to \infty} U_n$

$\lim_{n \to \infty} \infty + 1 = \infty$

Series diverges

a) $\{U_n\} = (-1)^{n+1}$

$\lim_{n \to \infty} (-1)^{n+1}$

$= (-1)^{\infty+1}$

$= (-1)^{\infty}$ $= +\infty$ for even

$-\infty$ for odd

this series diverges.

a) $\{U_n\} = \frac{n}{n+1}$

$\lim_{n \to \infty} U_n$

$= n \div n+1$

$= \frac{\cancel{n}}{\cancel{n}} \div \frac{\cancel{n}+1}{\cancel{n} \;\; n}$

$= 1 \div 1 + \frac{1}{n}$ $\therefore \frac{1}{n} = \frac{1}{\infty} = 0$

$= \frac{1}{1}$

$= 1$ (converges)

a) $\{U_n\} = 1 + (-\frac{1}{2})^n$

$\lim_{n \to \infty} U_n$

$= 1 + 0$ $\therefore$ for values less than 1 to the power of infinity $= 0$ $(1.2)^{\infty} = \infty$

$= 1$ (converges) $(0.8)^{\infty} = 0$

$(0.5)^{\infty} = 0$

Q) $\{U_n\} = \frac{(-1)^n}{n}$

$\lim_{n \to \infty} \frac{(-1)^n}{n}$

for odd terms

$= \frac{-1}{\infty}$ converges to 0

for even terms

$= \frac{1}{\infty}$ converges to 0

Q) $\sum_{k=1}^{\infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$

This Question onwards is from the P-series test .... i think

$\lim_{n \to \infty} = \frac{k^3 \left(3 - \frac{2}{k} + \frac{4}{k}\right)}{k^7 \left(1 - \frac{1}{k^4} + \frac{2}{k^7}\right)}$ $\frac{1}{\infty} = 0$

$= \frac{k^3 (3 - 0 + 0)}{k^7 (1 - 0 + 0)}$

$= \frac{3\cancel{k^3}}{k^{7\,4}} = \frac{3}{k^4} = \frac{1}{k^4}$ $\therefore \frac{1}{p^n}$ $n > 1$ converge, $n \leq 1$ diverge

= converge

$= \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2} \times k^4$

$= \frac{3k^7 - 2k^6 + 4k^4}{k^7 - k^3 + 2}$

$= \frac{3\cancel{k^7}}{\cancel{k^7}} = 3$; defined.

$\frac{1}{2} - 2$ $\quad 1.5 \quad -\frac{2}{5}$

$\frac{1-4}{2}$

Q) $\frac{5}{4} + \frac{7}{9} + \frac{9}{16} \ldots\ldots$

$n=1 \quad n=2 \quad n=3$

$$\sum_{n=1}^{\infty} \frac{2n+3}{(n+1)^2}$$

$$= \frac{2n+3}{n^2+2n+1}$$

$$V_n = \frac{2n}{n^2} = \frac{1}{n^1} \qquad \frac{1}{p^n} = \text{divergent}$$

$$= \frac{2n+3}{n^2+2n+1} \times n$$

$$= \frac{2n^2+3n}{n^2+2n+1}$$

$$= \frac{2n^2}{n^2} = 2$$

Q) $$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

$$= \frac{1}{n^2+1}$$

$$V_n = \frac{1}{n^2} \qquad \text{convergent}$$

$$= \frac{1}{n^2+1} \times n^2$$

$$= \frac{n^2}{n^2} = 1 \qquad \text{Defined}$$

e) $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{5 + n^5}$

$V_n = \frac{2n^2}{n^5} = \frac{2}{n^3} = \frac{1}{n^3}$ = converge

$= \frac{2n^2 + 3n}{5 + n^5} \times n^3$

$= \frac{2n^5}{n^5} = 2$ Deliver

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